

Exploring the energy system impacts of Nigeria's Nationally Determined Contributions and low-carbon transition to mid-century

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ABSTRACT

The 2015 Paris Agreement (COP21) mandated countries to submit their Intended Nationally Determined Contributions (INDC). In this paper, we develop the TIMES-Nigeria model and applied it to assess the impacts of Nigeria's NDC mitigation strategies and the ripple effects on the energy system. To this end, we modelled four mitigation scenarios in addition to the reference scenario. Increased renewables penetration, energy efficiency improvements, and fuel switching are identified as key mitigation strategies in the Nigerian NDC. Our results reveal profound restructuring of the Nigerian energy system with changes in the supply mix which mainly favour the deployment of solar PV. Our analysis shows that the residential sector has the largest potential for reducing energy demand while the power sector should be the focus for CO₂ emissions mitigation. Furthermore, we explore the impacts of carbon tax on the energy system. Our analysis discloses a significant improvement in energy security due to mitigation actions. We also shed lights on the local and international implications of the results. While our analysis should be counted as a preliminary study because of limited available country data, the outcomes are expected to provide useful insights with respect to low-carbon transition in the country.

1. Introduction

The IPCC Special Report on Global Warming of 1.5 °C shows that warming due to anthropogenic GHG emissions from the pre-industrial era to the present will remain for centuries, and will continue to cause further long-term changes in the climate system (IPCC, 2018). Climate change mitigation has become one of the most critical global policy challenges as it requires a coordinated international and local initiative (Dioha and Kumar, 2019). The challenge even becomes more overwhelming due to the complex interaction between the climate system and the energy system (Fragkos et al., 2017). In 2014, the 20th edition of the annual United Nations-led Conference of Parties (COP) invited all its members to prepare their Intended Nationally Determined Contributions (INDCs) which is aimed at reducing GHG emissions to a specific level by 2030. This was done in the context of adopting a legally binding instrument to cut down emissions by both developed and developing countries.

In December 2015, the 21st edition of the COP adopted the Paris Agreement which was reached by 195 parties and this constitutes a

historic milestone in the fight against climate change. The Paris Agreement covered key issues related to climate mitigation such as transparent accounting system for GHG emissions reduction, strengthening of country's adaptation capacities, and provision of finances for countries to achieve their INDCs.¹ In a nutshell, the Paris Agreement seeks to limit the global increase in average temperature to a level that is well below 2 °C in comparison with the pre-industrial levels and if possible, below 1.5 °C. In the run-up to COP21, a vast majority of countries, accounting for around 90–95% of global GHG emissions submitted their climate mitigation targets which are known as the INDCs. However, when a country officially ratifies its Paris Agreement, its INDC now becomes Nationally Determined Contribution (NDC).²

Nigeria is a lower middle-income developing country situated in the western part of Africa. Over the years, Nigeria has shown its commitment to the global fight against climate change. The country joined the United Nations Framework Convention on Climate Change (UNFCCC) in 1994 and ratified its Kyoto Protocol in 2004. Nigeria submitted its 1st and 2nd National Communications to the UNFCCC in 2003 and 2014 respectively, and has been hosting several Clean Development

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¹ <https://unfccc.int/process-and-meetings/conferences/past-conferences/paris-climate-change-conference-november-2015/cop-21>.

² We use the term NDC instead of INDC throughout this paper due to the fact that Nigeria has ratified its Paris Agreement and for consistency in nomenclature. However, we are aware that not all countries have yet ratified the Paris Agreement.

Mechanism projects as well (FGN, 2015). The country signed and ratified the Paris Agreement on 22nd September 2016 and 16th May 2017 respectively.³ In 2014, Nigeria became the largest economy in sub-Saharan Africa. However, despite this growth, significant development challenges remain. Poor access to modern forms of energy and high unemployment, amongst others, remain principal constraints on economic development. The human development index as of 2018 is still low at around 0.534 and a global rank of 158 out of 189 countries (UNDP, 2019). These challenges are of primary concern to the government and its vision is to grow the economy and reduce poverty for a more prosperous future for its citizens (FGN, 2015). The Nigerian NDC, therefore, focused on the delivery of direct development benefits and sustainable growth of the economy. Nigeria unconditionally pledged a 20% reduction on business-as-usual (BAU) emissions by 2030, and a 45% conditional commitment which can be achieved with financial assistance, technology transfer and capacity building (FGN, 2015). In order to achieve this ambitious target, different sectoral actions plans were developed. The sectoral action plans related to the energy sector are summarised in Table 1.

Achieving the Nigerian NDC will require transformational changes in its energy system. It will involve the change of current infrastructures and require decisions that will affect the lives of Nigerians in the next decades. Recent literature has also shown that ambitious climate mitigation policies imply profound energy system restructuring and robust policy ramifications (Riahi et al., 2015). Thus, investigating the profound energy system impacts of Nigeria NDC will improve the understanding of the challenges and opportunities facing the country in the context of its NDC. In this paper, we assess the energy-related implications of the climate mitigation strategies embedded in the Nigerian NDC in terms of projected changes in the energy supply system, energy demand, associated CO₂ emissions, anticipated investment in energy technologies, as well as the co-benefits. We proceeded to look at the ripple effects if the NDC strategies are maintained up to 2050. To this end, we develop a TIMES (The Integrated MARKAL-EFOM System) model for the Nigerian energy system. We build and simulate a BAU scenario where it is assumed that there is no significant effort towards decarbonising the Nigerian energy system. We then develop and simulate alternative NDC scenarios based on the proposed decarbonisation strategies. The results of our analysis provide evidence on the changes required in the energy system of Nigeria towards achieving its NDC targets.

The remainder of this paper is structured as follows. Section 2 presents a brief review of the literature on energy system modelling and

different modelling studies conducted globally on NDCs. Section 3 describes the methodological approach based on the TIMES modelling framework and introduces the TIMES-Nigeria (TINIG) model which was used to carry out this study. Section 3 also presents the development of the different scenarios modelled in this study. Section 4 presents the results and analysis; comparing the different mitigation scenarios in terms of impacts on the energy system as well as emissions. Section 5 draws some conclusions, discussing the importance of the study and the main recommendations for policymakers.

2. Previous works and study relevance

2.1. Previous works

Energy system models are tools which are used to examine the energy, economic, and environmental impacts of different energy/climate policies in order to inform policy decisions (Dioha et al., 2019). While many classifications exist, the models are generally classified as bottom-up and top-down models (Herbst et al., 2012). The bottom-up models have a high degree of technological representation, but they do not usually account for the macro-economic implications of climate or energy policies. In contrast to bottom-up models, the top-down models represent the economy as a whole on a national or regional scale and evaluate the aggregated impacts of climate or energy policies in monetary values, but they lack detail representation of technologies. For a detailed description of energy system models, see the work of Dioha (2017).

In recent years, studies examining low-carbon development pathways with the application of different energy system models have increased significantly due to the increased pressure faced by policymakers on the need to transition to a low-carbon economy. Here, we took a snapshot of a few. At the global level, the International Energy Agency (IEA) conducts modelling studies on GHG mitigation scenarios and the role of technology innovations in catalysing the process (IEA, 2017). Ekholm et al. (2008) used the TIAM-WORLD model to analyse mitigation costs for different scenarios in different regions of the world. In a different study at a regional scale, TIAM-ECN model was used to examine the prospects for large-scale use of low-carbon energy technologies in Africa (van der Zwaan et al., 2018). At a national level, Chiodi et al. (2013) used the Irish-TIMES model to evaluate the energy system impacts of the European 2050 GHG mitigation targets on Ireland's energy system. Many detailed studies on low-carbon development around the world using energy system models have been undertaken and they are summarised in Gargiulo and Gallachóir (2013).

As regards the NDCs, a large body of literature has evaluated the implications of the NDCs. Hof et al. (2017) used the IMAGE integrated assessment model (IAM) to examine the annual abatement costs of realising the NDC mitigation targets, as well as the additional costs if countries align their targets with keeping global warming well below 2 °C and “pursue efforts” towards 1.5 °C. Qi and Weng (2016) analysed the implications of a linked emissions trading systems in achieving the NDC emission reduction target by 2030 using a multi-regional Computable General Equilibrium (CGE) model. Their study suggested that global emissions trading systems reduce the emission reduction cost in Annex I countries and that a market-based emission trading system for achieving the NDC is cost-effective. In a different study, Fragkos et al. (2018) coupled global and national models to evaluate the policy implications of NDCs. Similarly, Vandyck et al. (2016) combined a technology-rich bottom-up model with a top-down CGE model to investigate the techno-economic implications of the global NDCs and reports that energy demand reduction and a decarbonisation of the power sector are essential for emission reductions up to 2050.

On the global front, Fragkos and Kouvaritakis (2018) investigated CO₂ emission trajectories and energy system implications for the Reference, NDC, and cost-optimal 2 °C mitigation pathways globally and in 4 major economies (EU, USA, China, India), using the PROMETHEUS

Table 1
Sectoral action plans for Nigeria NDC.

Sector	Mitigation strategies in the Nigerian NDC
Power	13 GW of decentralised solar PV, more efficient gas-fired power generation, reduction of electricity transmission and distribution losses.
Residential	Upscaling household electricity access, energy efficiency improvement of electrical appliances (e.g. light bulbs, air conditioners, refrigerators etc.), use of improved cookstoves, and encouraging energy saving attitude.
Transport	Modal shift in passenger and freight transport, fuel switch (from gasoline and diesel to CNG) and improving overall vehicle efficiency.
Industry	Improving the energy efficiency of industrial processes and fuel switching to less carbon-intensive fuels.
Commercial	Enhancing penetration of energy efficient technologies (e.g. light bulbs, air conditioners, refrigerators etc.), and encouraging energy saving attitude.

³ https://treaties.un.org/Pages/ViewDetails.aspx?src=TREATY&mtid_g_no=XXVII-7-d&chapter=27&clang=en.

global energy-economy model. In a similar study, Iyer et al. (2015) x-rayed the long-term energy and economic implications of NDCs using the GCAM model and suggested that NDCs have the effect of reducing premature retirements of fossil fuel power plants. However, in the context of strengthened NDCs, as encapsulated in the Paris Accord, Aldy et al. (2017) proposed robust frameworks to compare different countries mitigation efforts while Hühne et al. (2018) used a large variety of evaluation methods to assess climate proposal ambitions. In a different research, Sanderson et al. (2016) reviewed potential global climate trajectories and proposed a set of idealised emission pathways which are consistent with the Paris temperature targets. With respect to GHG emissions, Benveniste et al. (2018) provided an analysis of uncertainty sources and their implications on global GHG emissions by 2030 with reference to the full achievement of the NDCs, and suggested that NDCs significantly shift GHG emissions towards developing countries.

At the regional level, Fragkos et al. (2017) reviewed and assessed the energy system impacts of the European NDC as well as low-carbon transition pathway to 2050 using the PRIMES and GEM-E3 models. The study showed that the European Union NDC will have around 0.4% impact on the GDP by 2030 in comparison with the reference scenario. In Asia, Zhai et al. (2018) explored the energy system impacts of the NDCs of Asian Development Bank developing country members. The study reports that renewable energy will be the largest contributor to the increase in electricity production under the NDCs and around US\$4.8 trillion will be needed for implementing the NDCs of the Bank's developing country members.

Shifting to national level assessments, there have been different country-level scenario analysis of the NDCs which are aimed at informing low-carbon decisions. Some of these studies include Mu et al. (2018) who analysed the economic implications of achieving China's NDC using the Chinese CGE model. The analysis indicates that the production and prices of coal and electricity will be the most affected by China's NDC. Akimoto et al. (2015) analysed the economic and energy implications of Japan's NDC development strategies. Spencer and Pierfederici (2015) based their analysis on national-level models to provide insights on the energy system impacts of major-carbon emitting countries' NDCs. In another study, van Soest et al. (2017) employed IAMs to evaluate the emission trajectories and the energy system impacts of 11 major economies for a reference and optimised 450 ppm CO₂ eq mitigation pathways and compare the results with the NDCs. They found that in South Korea, Canada, Brazil, European Union, USA, and Mexico (conditional target), the emissions reduction targets of their NDCs are closer to the mitigation needed for the 2 °C pathway, but further suggests that there is still a large gap in other countries.

The reviewed studies show the advancement of knowledge boundaries in assessing the impacts of NDCs in various parts of the world. However, we identified certain research gaps which when addressed can further improve our understanding of the energy system implications of NDCs especially, from the perspective of developing countries. From the literature, we found that many studies on the NDCs have been on a global scale and few national case studies exist. However, NDCs were developed at the country-level and studying their implications at the country-level could contribute to informing national low-carbon strategies. The reviewed literature shows that only a few studies have analysed the implications of the NDCs beyond 2030. Therefore, it is important to understand the impacts of the NDCs even beyond 2030 since the NDCs were actually designed to avert long-term climate crisis and having an insight on the long-term implications of the NDCs can provide the basis for strengthening the NDCs. Most studies examining the impact of the NDCs mainly focus on the energy, GHG emission, and economic implications without considering the co-benefits. However, understanding the co-benefits of the NDCs is very much important in the context of sustainable development. Finally, most reviewed studies on quantitative analysis of the NDCs focussed on developed countries in North America and Europe, while just a few have focussed on developing countries, and none so far, for a country in Africa. This is

comprehensible as the developed countries are the major historical GHG emitters, but experience has shown that without significant emission reduction from developing countries, the goal of the Paris Accord may not be achieved. Hence analysis of the NDCs needs to be extended to the developing countries such as Nigeria which are envisaged to have higher emissions in the future due to the yearning for socio-economic development.

2.2. Study relevance

Given the research gaps identified, this paper adds to the body of literature in this domain by first introducing the TINIG energy system model—the first of its kind for Nigeria. The paper, in a useful manner, improves the understanding of the energy system restructuring due to the Nigerian NDC at the sectoral levels, and its policy implications. The modelling assessment also extends up to 2050 in order to provide long-term perspectives on the NDC. The paper not only investigates the energy system implications of the Nigerian NDC, but it goes a step further to analyse the co-benefits and the impacts of a carbon tax. While the current study analysed the NDC using a national lens, it is worthwhile to note that the Paris Agreement is an on-going global issue and the present study is therefore of wider importance to other governments and international agencies. Additionally, beyond the mitigation scenarios analysed here, the developed TINIG model can be used to examine other different kinds of low-carbon pathways. The study outcome generally adds to the literature on low-carbon development pathways as well as provides support to policymakers to identify focus areas for climate policy in the future.

3. Methodology

3.1. The TIMES energy system model

We use the TIMES model generator to build an economy-wide energy system model for Nigeria. TIMES is a widely used bottom-up optimisation modelling tool, initiated by the Energy Technology Systems Analysis Programme in an implementing agreement with the IEA (Chiodi et al., 2013). The TIMES model generator is currently used in over 70 countries for different techno-economic studies relating to energy and climate policies. The model combines all the advanced characteristics of MARKAL models and to a lesser extent that of EFOM models. TIMES is usually used to model the entire energy system of a country. However, it can also be used for detailed sectoral analysis. At a country level, a typical TIMES model includes energy extraction, conversion, distribution, and end uses. Depending on the region, a TIMES model can also contain various forms of energy trade. Each stage in a TIMES model is described by means of specific technologies which are characterised by their techno-economic parameters. The model also accounts for environmental emissions from fossil fuel combustion and industrial processes.

TIMES is a dynamic linear programming model which assumes that energy markets are under perfect competition. The objective function of TIMES (Equation (1)) is to minimise the total discounted cost of the energy system while adhering to the many technical and environmental constraints defined in the model. This is equivalent to maximising the net total surplus (i.e. the sum of consumers' and producers' surpluses). The main outputs of a TIMES model include demand and supply-side technology deployment, resource-wise final energy requirements, future investments in technologies as well as environmental emissions at each period of time.

$$NPV = \sum_{r=1}^R \sum_{y \in YEARS} (1 + d_{r,y})^{REFYR-y} \times ANNCOST(r, y) \quad (1)$$

where NPV is the objective function of TIMES (i.e. the net present value of the total cost of all regions); R the number of regions in the study area;

YEARS the set of years in the model for which there are costs; $d_{r,y}$ the discount rate; REFYR the reference year for discounting; and ANNCOST (r,y) the total annual cost in region r and year y .

The advantages of TIMES in comparison to other models include its integrated framework for analysing resources flow from extraction to end-use point; detailed technological representation; imposition of user-defined constraints; ability to run numerous scenarios etc. To successfully build a TIMES model, numerous kinds of data are required as TIMES is a data-intensive model. The user is required to provide the following category of data: global assumptions, energy service demands, energy carriers, energy demand and supply technologies, and environmental emission factors. The full technical documentation of the TIMES model is given in Loulou et al. (2016). The TIMES model is widely used internationally, and thus, the results of the model can be compared with other countries.

3.2. The TIMES-Nigeria (TINIG) model

In this paper, the TINIG model has been developed to analyse a range of mitigation scenarios in order to inform policy decisions. The model represents the energy system of Nigeria and its possible long-term development. TINIG is a single region model which spans 40-years (2010–2050), with shorter time intervals of 5 years. All costs defined in TINIG are in 2010 United States dollars (US\$). The current study considered 10% as the annual discount rate. This high discount rate is used owing to the fact that Nigeria is a developing country in dire need of infrastructural projects, thus, it is only reasonable to use a higher discount rate to determine the present value of future cash flows. Moreover, an earlier study on Nigeria's energy and emissions outlook by the ECN also used a 10% discount rate (ECN, 2015).

TINIG model basically consists of the demand and supply components. The demand component of the model consists of exogenous energy service demands which were specified in the base year (2010) and subsequent milestone years up to 2050. Depending on the country and the availability of data, the number of energy service demands can vary. In TINIG model, different econometric and engineering methods were used to estimate the energy service demands. The demand component of the model consists of 30 different energy services. They include 12 for the residential sector, 8 for transport, 5 for industry, 4 for commercial, and 1 for agriculture. In the residential sector, we split households into rural and urban. Energy services considered at the household level include lighting, cooking, water heating, cooling etc. In the transport sector, we considered only land transport due to data limitation and we classified it into passenger and freight transport systems. While air and water transport are also important, land transport in Nigeria consumes over 90% of energy in the transport sector (Chukwu et al., 2015), and thus gives us a true representation of energy use in the sector. In the industry sector, energy service demands were estimated in terms of industrial output for cement, steel, fertilizer, and paper industries while other industries were estimated in terms of final energy consumption. For the commercial sector, similar energy services in the residential sector were considered while aggregated final energy was used for the agriculture sector as it consumes less than 1% of final energy demand in the country (ECN, 2013). Accordingly, to deliver the aforementioned energy service demands, each energy demand sector is defined by a database of technology which includes data on the technical, economic, and environmental characteristics of the technologies.

The supply component of TINIG model consists of the primary energy sources in the country, their potentials, prices, and future prospects. The energy sources consist of non-renewable and renewable sources of energy. The non-renewable energy sources modelled include crude oil, natural gas, coal, and nuclear while the renewable sources include solar, wind, hydro, and biomass. In the transformation sub-component, electricity production was also split into grid-connected and off-grid systems. Electricity production sources/technologies modelled in TINIG include hydro (large and small), natural gas,

bioenergy, coal, nuclear, solar, and wind. However, electricity import was also modelled but have not been activated in the present model. Other secondary energies encompass refinery products and liquid bio-fuels. Although Nigeria has four established oil refineries, their cumulative capacity is just enough to satisfy only around 25% of Nigeria's oil products demand and thus, the country imports oil products to augment the shortfall in local production (Alaba and Agbalajobi, 2014). Consequently, the importation of oil products was also modelled in TINIG. Due to word limit constraint as per the journal (*Energy Policy*) rule, the detailed technical report on the development of TINIG model (assumptions, data, and methodologies) is provided in the supplementary information (Appendix A). Fig. 1 shows the simplified version of TINIG model.

3.3. Scenarios description

Energy systems are complex and dynamic and are often not well understood. Usually, scenarios are the best tool for understanding different energy development pathways and their implications. In scientific energy studies, scenarios are described as internally consistent, replicable set of storylines or theories about the important relationships and driving factors of change, which are usually gotten from the understanding of both history and the current situation (Kram et al., 2000). In this paper, our scenarios are plausible descriptions of how the Nigerian NDC may develop based on an internally consistent and coherent set of assumptions on key socio-economic driving forces and relationships. It is worth mentioning that the scenarios are just images of alternative futures and they neither predict nor forecast the future and also, we do not hold such a single view. Our aim is to quantitatively assess the impacts and to generate insights on how the Nigerian energy system will respond to alternative low-carbon pathways. To this end, we use the Schwartz scenario planning process in building our scenarios which are based on his book titled '*The Art of the Long view: planning for the future in an uncertain world*' (Schwartz, 1991). The Schwartz scenario planning approach has been employed in several energy/climate studies (Awopone et al., 2017; Emodi et al., 2019; Gamas et al., 2015; Ghanadan and Koomey, 2005; McPherson and Karney, 2014; Ouedraogo, 2017). The steps involved in using Schwartz's methodology is shown in Fig. 2. The detailed description of these steps is given in Pastor (2009).

All in a bid to ensure that our scenarios are valid, reliable, and applicable, we conducted a series of informal interviews with relevant stakeholders and opinion leaders in the Nigerian energy/climate landscape. This was done to disclose our initial scenario estimates and to get the best information from authoritative sources as well as ensure that our scenarios are in line with current trends and future expectations of the Nigerian energy system in terms of its NDC. In total, 14 different stakeholders were interviewed. The stakeholders involve those with government agencies, civil societies, and academia. This has been done to get a balanced view of the subject matter. Sample questions of the interview are given in the supplementary document in Appendix A. Affiliations of the stakeholders include: Energy Commission of Nigeria; Federal Ministry of Environment; Federal Ministry of Transport; Federal Ministry of Power, Works & Housing; Federal Ministry of Industry, Trade & Investments; Nigerian National Petroleum Corporation; Manufacturers Association of Nigeria; Nigerian Alliance for Clean Cookstoves; Federal University of Technology Minna; and University of Abuja. For the purpose of this study, 5 scenarios have been developed and they are summarised as follows:

Business-as-usual (BAU): This scenario is the most basic task in energy system analysis as it provides the benchmark for comparison with alternative policy scenarios. TIMES—an optimisation model gives the cost-optimal energy system configuration, and this may not be the reality in the Nigerian energy system as choices of energy technologies is not always determined by cost. Thus, we assume a frozen scenario in which only the existing technologies as in the base year are available for the model to select from 2010 to 2050 in all sectors of the economy. To

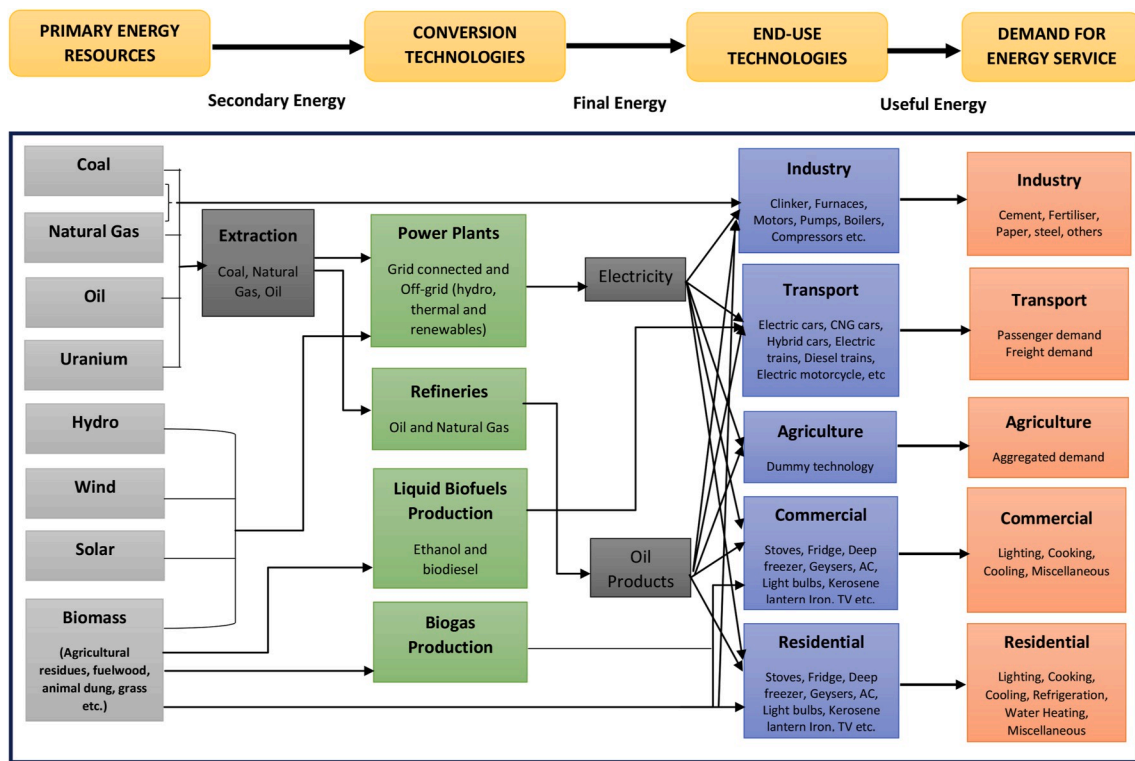


Fig. 1. Simplified representation of the TINIG reference energy system.

this end, we embedded several constraints in the model to improve the realism associated with the future energy pathways of Nigeria. This is also necessary because of the intrinsic nature of linear programming models like TIMES, which could otherwise in several cases deliver extreme technology switches (Chiodi et al., 2013).

Unconditional NDC (UCD): Under this scenario, it is assumed that Nigeria will not receive any international support in achieving its NDC. This implies that all activities/projects outlined in the sectoral action plans of the NDC will be achieved internally. Thus, this scenario is built on the premise that projects funding is sourced locally, and the government also takes the realisation of the Nigerian NDC seriously. This scenario emphasizes that policy decisions and investments for realising the NDC must be made in the next 5–10 years in order to meet up with the deadline of the Paris Agreement. In this scenario, energy efficiency is gradually adopted across all sectors; there is fast installation of the 13 GW decentralised solar PV, and gradual penetration of low-carbon fuels in the transport sector as well as smart transmission and distribution networks.

Conditional 1 NDC (CD1): Our engagements with relevant stakeholders in the Nigerian energy and climate landscape shows that there are many uncertainties with respect to international support for the NDC. CD1 is a pessimistic scenario in which international support will be received in terms of funding and technology transfer but at a limited level. In comparison with the UCD scenario, this scenario is more ambitious, and it is based on moving all sectors towards their technological frontier within limited international economic support received. On the demand side, there are moderate improvements in energy efficiency across all sectors while in the supply side, there is a moderate expansion of decentralised solar PV capacity and transmissions and distribution losses in comparison to the UCD scenario.

Conditional 2 NDC (CD2): This scenario is based on an optimistic future in which there is unprecedented international support from bilateral/donor agencies. The scenario explores ways in which low-carbon technologies will be diffused massively across all sectors of the economy at a higher rate in comparison with the CD1 scenario. It

encompasses better uses of transforming technologies and fuels in all sectors, while also maintaining a low-carbon lifestyle in terms of transport choices. Cities are well equipped with the type of infrastructures required to reduce travel km, LPG cooking equipment is greatly adopted in the residential and commercial sectors. The industrial sector energy intensity reduces significantly by making use of more efficient processes, and by increasing the share of natural gas in the industrial fuel mix. Alternative power technologies, specifically, decentralised solar PV capacity is also increased.

Emission tax (ETX): As an experiment, we examine the impact of carbon pricing on the Nigerian energy system and CO₂ emissions from the energy sector. We also analyse the technologies that are cost-optimal under the carbon price policy. Carbon pricing has become one of the essential climate mitigation policy tools. Recently, many countries are showing interest in implementing carbon pricing policy but the vast majority of global emissions today are still not under any price (Carbon Market Watch, 2017). Moreover, even the countries that have carbon pricing policies in place keep the prices at very low levels which are not high enough to make a significant contribution towards reaching the goals of the Paris Agreement (Carbon Market Watch, 2017). The Paris Agreement provides the overall global framework for the fight against climate change. However, carbon pricing was not specifically mentioned in the Paris Agreement, but it is an essential policy tool that many countries are now using to reduce national GHG emissions in order to help reach their international climate commitments such as the NDCs. Around the world, more and more governments are introducing different carbon pricing schemes, but so far, most prices have been below US\$10/tCO₂. Though it is very difficult to accurately price the damages attributed to climate change (e.g. sea level rise, spread of tropical diseases, biodiversity loss, ocean acidification, drought, famine, and political instability), the High-Level Commission on Carbon Prices has found that a price of around US\$40–80/tCO₂ by 2020 and US\$50–100/tCO₂ by 2030 is required to achieve the Paris Agreement (Carbon Market Watch, 2017). Accordingly, in this scenario, we modelled a carbon tax in accordance with the prices proposed by the

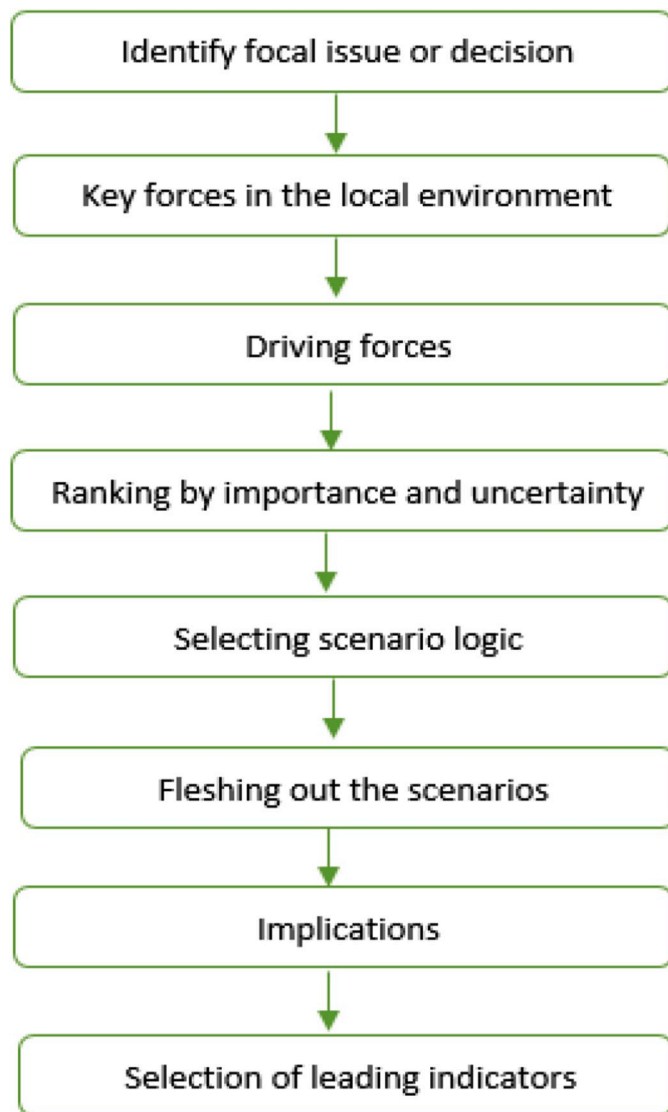


Fig. 2. The 8 steps involved in Schwartz's scenario planning process.

High-Level Commission on Carbon Prices and the model is allowed to run freely.

To put the foregoing words into numbers, we have provided the detailed characterisation of the scenarios in the supplementary information (Appendix A). It contains the specific penetration rates of different assumptions, energy infrastructures and technologies across the various sectors in all the scenarios considered. It is worthwhile to mention that all demographic and macro-economic projections used in this study were kept constant across all scenarios.

3.4. Co-benefits

The co-benefits of climate mitigation is an area that is still largely unexplored. Thus, it is pertinent to look at the big picture of low-carbon transition by evaluating simultaneously the co-benefits of this transition. Going beyond finances, another reason why low-carbon development is at a slow pace in Nigeria is the lack of information by policymakers on the multiple benefits low-carbon development can potentially deliver. For any effective policy reform, these co-benefits need to be quantified and communicated appropriately. While transitioning to a low-carbon economy in Nigeria may incur more investments; if the co-benefits are considered, the true cost may be lower than the initial investment estimated. Thus, it is only important that researchers and policymakers

understand this dimension in energy transition analysis. Moreover, the Organisation for Economic Cooperation and Development (OECD) has opined that the benefits of climate change mitigation policies in the coming decades would be mainly the co-benefits since the direct benefits of mitigation policies are expected to bear fruits in the longer-term (Johannes Bollen et al., 2009). In this paper, we pay particular attention to energy security.

Energy security is defined by the IEA as 'the uninterrupted availability of energy sources at an affordable prices'.⁴ Energy security is a serious geopolitical issue and there have been dozens of articles published over the years on this concept and the issues pertaining to it (Cherp and Jewell, 2014). In recent years, energy security has even become more important as an adequate supply of energy is needed to drive the economic development of countries. While analysing energy security, recent studies have proposed a set of indices which can be used in examining the energy security of a country. In this paper, we analyse energy security through the lenses of energy intensity (EI) and energy diversity (ED) (Selvakkumaran and Limmeechokchai, 2013). The EI is used to assess the penetration of energy efficient technologies in the economy and it is defined as the total energy used per unit of GDP. The ED index is used to assess the significance of diversification in terms of abundance and equitability of energy resources (Chunark et al., 2015). It is worthwhile to mention that the ED index does not capture fluctuations in the supply of renewable energies but only shows the diversity of energy sources in the supply mix. In this paper, the ED index is analysed using the Shannon Index H' , which is a diversity index used to measure biodiversity (APEREC, 2007). The Shannon Index is given in Equation (2), while Equation (3) has been used to determine the ED index for all scenarios.

$$H' = - \sum_{i=1}^R p_i \ln p_i \quad (2)$$

where p_i is the number of characters belonging to the i th of energy supply in the country.

$$ED = \frac{H'}{\ln T} \quad (3)$$

where T is the number of energy sources which are utilised in the country.

4. Model results and analysis

This section reports in detail the model results for the BAU scenario. The results for the UCD, CD1, CD2, and ETX scenarios have been provided only for the year 2050 for comparative analysis amongst the scenarios.

4.1. Primary energy supply

Fig. 3 shows the results of the primary energy supply in the BAU and the decarbonisation scenarios. In the BAU scenario, the total primary energy supply reaches 6940 PJ by 2030 and 17239 PJ by 2050 from 2471 PJ in 2010. This indicates an increase by around 6 times in 40 years. Biomass fuelwood continues to remain the dominant fuel with its share gradually dropping from 66.7% in 2010 to around 41.5% by 2050, and supply from 1648 PJ to 7155 PJ. The share of oil in the energy supply mix grows from 21.3% in 2010 to 36.8% by 2050. Even though the magnitude of coal in the supply mix increases from 3 PJ in 2010 to about 34 PJ in 2050, its share remains the lowest at around 0.2%. Thus, by 2050 in the BAU scenario, 58.1% of the primary energy is expected to come from fossil fuels (coal, oil, and gas) while renewable hydro and biomass will account for the remaining 41.9%.

⁴ <https://www.iea.org/topics/energysecurity/>.

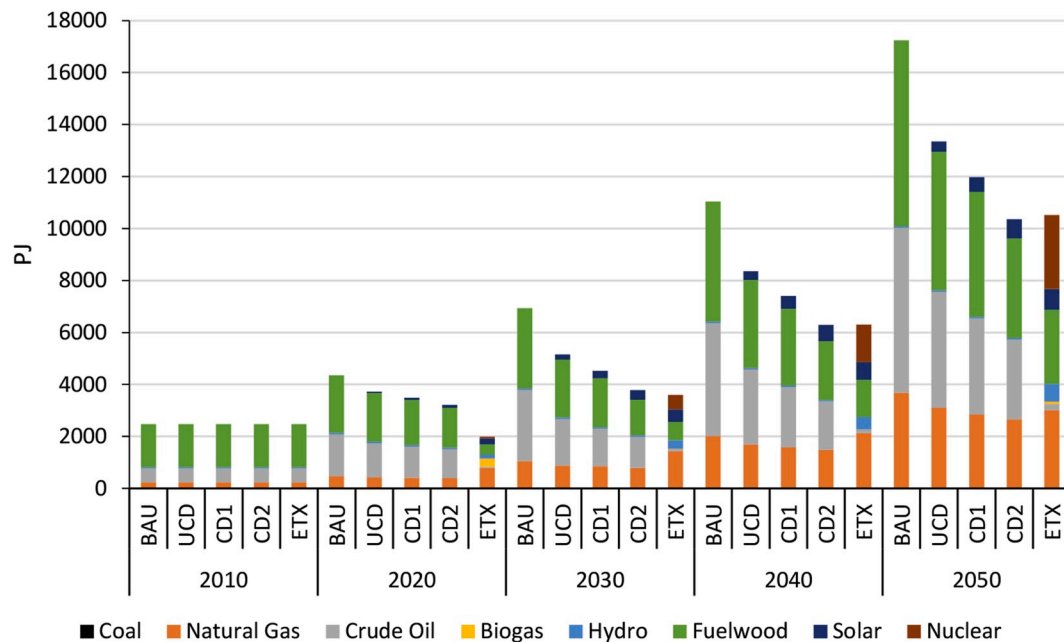


Fig. 3. Primary energy supply by source.

Nevertheless, in our alternative scenarios, we see a fall in the share of fossil fuels in the primary energy supply mix. In UCD, CD1, CD2, and ETX scenarios, total primary energy supply is seen to fall respectively, by 22.4%, 30.5%, 39.9%, and 39.0% when compared to the BAU scenario. The general decrease in primary energy supply observed across the alternative scenarios can be attributed to the reduction in the consumption of oil and fuelwood by 2050. These reductions are achieved on the account of high-efficiency measures taken across sectors. In comparison with the BAU case, the use of oil in primary energy supply reduces by 1876, 2648, 3279, and 6111 PJ in UCD, CD1, CD2, and ETX scenarios respectively, owing to the lower demand of oil products in the transport sector. The use of fuelwood in primary energy supply is also observed to reduce respectively, by 1830, 2349, 3318, and 4297 PJ in UCD, CD1, CD2, and ETX scenarios; a drop recorded due to higher penetration of efficient cookstoves in rural and urban households. An interesting part of the energy supply results is the ETX scenario. It may be recalled that all constraints were removed from the model in this scenario and it was allowed to operate as a full optimisation model under a carbon tax regime. Our analysis shows that nuclear energy will

account for around 27% of the primary energy supply mix in this scenario. A significant value which is due to the selection of nuclear power technology as one of the cost-optimal technologies under a carbon tax.

4.2. Electricity production

The electricity generation fuel mix is given in Fig. 4. In BAU scenario, total power generation increases from 72 TWh in 2010 to 208 and 671 TWh in 2030 and 2050 respectively (equivalent to a compound annual growth rate (CAGR) of 2.7% and 5.8% in 2030 and 2050 respectively). The share of diesel and gasoline in total power generation is 44.2% in 2010 and this gradually drops to 42.7% by 2050. This is because of the gradual improvements in the centralised power supply system which now offsets the use of off-grid diesel & gasoline generators. Due to the massive investment in thermal power plants, the share of natural gas (CCT & SCT) in generation rises from 32.6% in 2010 to 54.9% in 2050. However, the share of renewable continues to remain insignificant in the power generation mix.

In UCD, CD1, CD2, and ETX scenarios, the requirements for low-

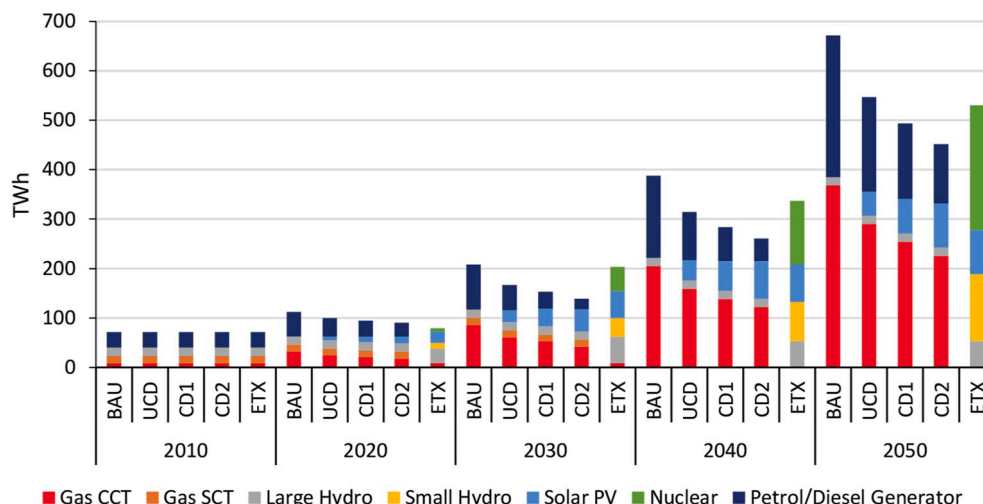


Fig. 4. Electricity generation by source.

carbon and efficient electricity are increased considerably. Power generation is seen to drop to 547, 493, 452, and 530 TWh in 2050, which reflects a saving of 18.6%, 26.5%, 32.7%, and 21.0% respectively, when compared with the BAU scenario. We see a drop in the share of fossil fuels in the alternative scenarios by 2050 which is mainly due to the development of solar PV generation in the country. Diesel and gasoline generation contribute around 35.1%, 30.8%, and 26.6% in UCD, CD1, and CD2 scenarios respectively. Despite the investments in solar PV technologies, our analysis shows that solar PV generation will contribute only about 8.8%, 14.4%, 19.8% in UCD, CD1, and CD2 scenarios respectively; whereas, natural gas will continue to dominate the generation mix with about 50% contributions in all scenarios. However, in the ETX scenario, natural gas and diesel & gasoline are totally flushed out of the generation system by 2050, and nuclear, solar PV, small and large hydro are seen to account for 47.5%, 16.8%, 25.5%, and 10.2% respectively, to the power generation mix.

4.3. Final energy demand

Total final energy demand projections for each scenario is presented in Fig. 5. Final energy demand in BAU scenario grows from 2524 PJ in 2010–5866 PJ in 2030 and 13896 PJ in 2050, increasing by about 4.5 times in 40 years. In the mitigation scenarios, due to several demand management measures and fuel efficiency enhancement in the demand technologies, the total final energy demand is expected to decline respectively, by 22.4%, 31.3%, 39.5%, and 45.5% in UCD, CD1, CD2, and ETX scenarios by 2050. Thus, the present result helps us understand the important role of energy efficiency in ensuring an energy secure future.

With respect to the demand sectors (Fig. 6), the residential sector percentage share in final energy demand drops from 66% in 2010 to around 30%, 27%, 26%, 23%, and 13% in BAU, UCD, CD1, CD2, and ETX scenarios respectively, by 2050. Furthermore, we see an onward increase in the share of the industry sector in final energy demand from 12% in 2010 to around 42%, 43%, 44%, 47%, and 53% in BAU, UCD, CD1, CD2, and ETX scenarios respectively, by 2050. The respective decrease and increase of residential and industrial sector contribution in final energy consumption over time are due to the magnitude of consumption in these sectors. However, the shares of transport and commercial sectors were relatively stable in the ranges of 18–23% and 10–12% respectively, in all scenarios over the modelling timeframe. At

the sectoral level, reductions in final energy demand are mainly evident in the residential sector.

Our results further suggest that per capita energy consumption (Fig. 7) will increase significantly across all scenarios when compared to the base year value. This is because of the projected increase in population growth and economic activities. As earlier noted, there is serious yearning for socio-economic development in the country. In the BAU scenario, per capita energy demand grows from 16 GJ/person in the base year to about 35 GJ/person in 2050. However, our analysis shows that per capita energy demand by 2050 in UCD, CD1, CD2, and ETX scenarios will drop to 27, 24, 21, and 19 GJ/person respectively. It is also worth mentioning that currently, there is no direct feedback between the TINIG model results and the economy. Hence, all scenarios assume that GDP grows at an average of 7% per annum over the modelling horizon.

4.3.1. Residential sector

Fig. 8 illustrates the final energy demand of the residential sector over the modelling period for each scenario. Due to growth in per capita income and number of households, total final energy demand of this sector (rural and urban) is expected to grow by 152% in the BAU scenario, from 1675 to 4228 PJ in 2010 and 2050 respectively. The rural households account for around 68% of the final energy demand in 2010 (due to widespread use of inefficient biomass cookstoves). However, in 2050, this share declined to 36% as a result of massive urbanisation, which, in turn, reduces the energy demand of the rural households. In UCD, CD1, CD2, and ETX scenarios, the sector's total final energy demand declines respectively, by around 30.1%, 42.3%, 55.1%, and 77% in comparison to the BAU case. Another important difference between the BAU and the mitigation scenarios is the percentage share of fuels in the residential sector. Biomass fuelwood continues to be a dominant source of energy in both rural and urban households as it is used for cooking and heating in exception of the ETX scenario where electricity and biogas cookstoves are selected for cooking purpose. In BAU scenario, fuelwood consumption in rural and urban households increases from 974 and 382 PJ in 2010 to about 1148 and 1651 PJ in 2050 respectively. Due to the penetration of efficient cookstoves in the sector, the consumption of fuelwood decreases in the ranges 32–70% (rural) and 28–64% (urban) across UCD, CD1, and CD2 scenarios. Improved biomass cookstoves are up to 5 times more efficient than the traditional biomass cookstoves.

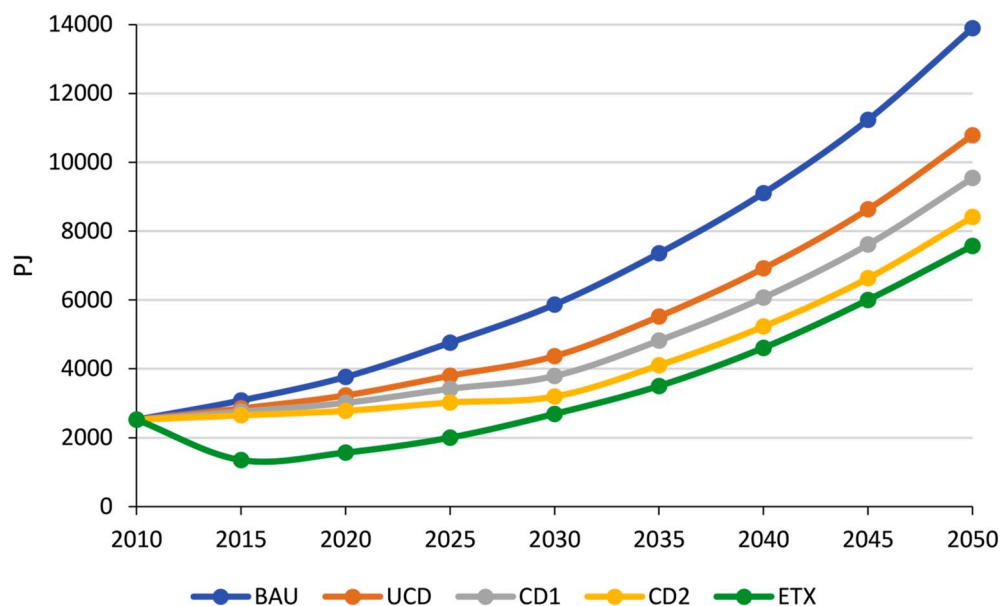


Fig. 5. Total final energy demand projections by scenario.

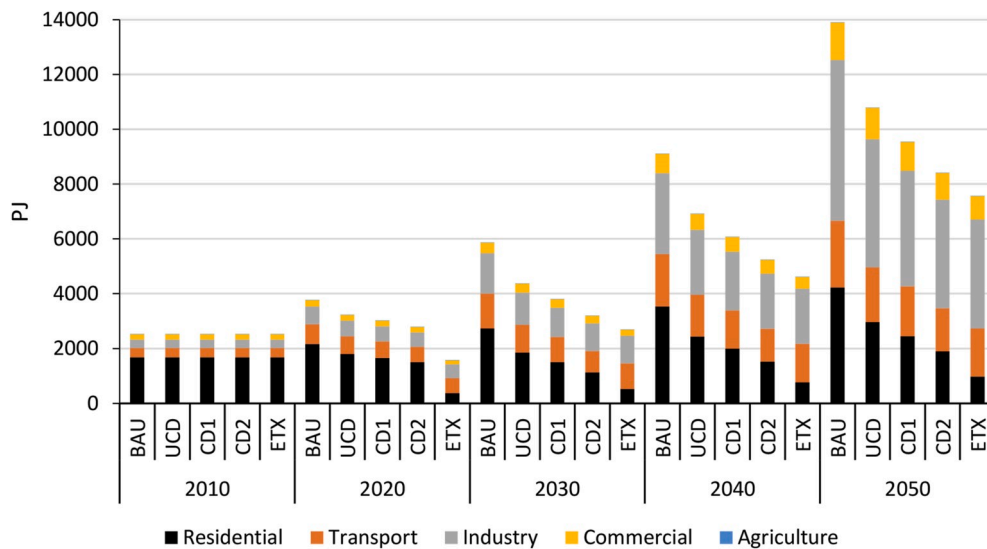


Fig. 6. Sectoral decomposition of total final energy demand.

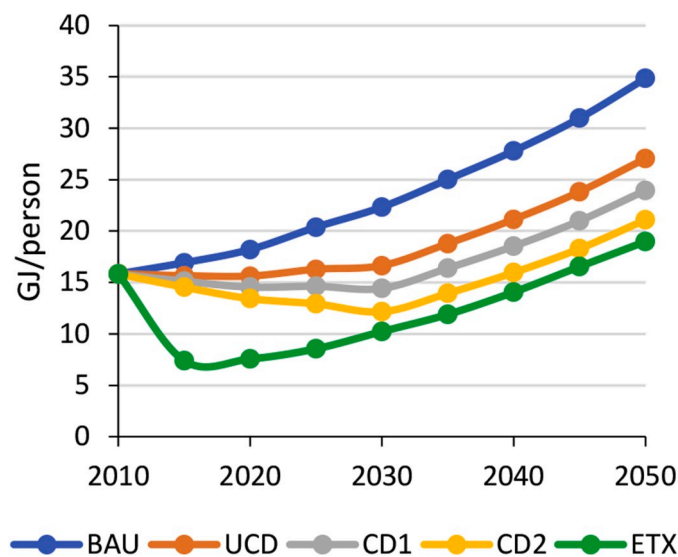


Fig. 7. Projections of per capita energy consumption by scenario.

Furthermore, owing to greater appliance penetration and electrification, the electricity consumed by the sector increases in the BAU scenario from 20 to 142 PJ (rural) and 57–634 PJ (urban) in 2010 and 2050 respectively. As expected, with the implementation of low-carbon strategies such as efficient air conditioners and refrigerators in the alternative scenarios, electricity consumption in the sector drops in the ranges 14–21% (rural) and 16–25% (urban) across UCD, CD1, and CD2 scenarios. Surprisingly, our analysis shows that electricity consumption in the sector increased by around 45% and 8% in rural and urban households respectively. This is because, in the ETX scenario, almost all end-use is electrified. Induction stove and biogas stoves displace LPG stove. Solar energy technologies are also adopted for water heating purpose.

4.3.2. Transport sector

Moving to transport, Fig. 9 illustrates energy consumption in the transport sector over the modelling period for each scenario. Transport final energy demand is expected to grow at a CAGR of 4.7% from 346 PJ in 2010–2439 PJ in 2050. This corresponds to around 6 times increase in 40 years. The sizeable growth in the transport sector can be attributed to

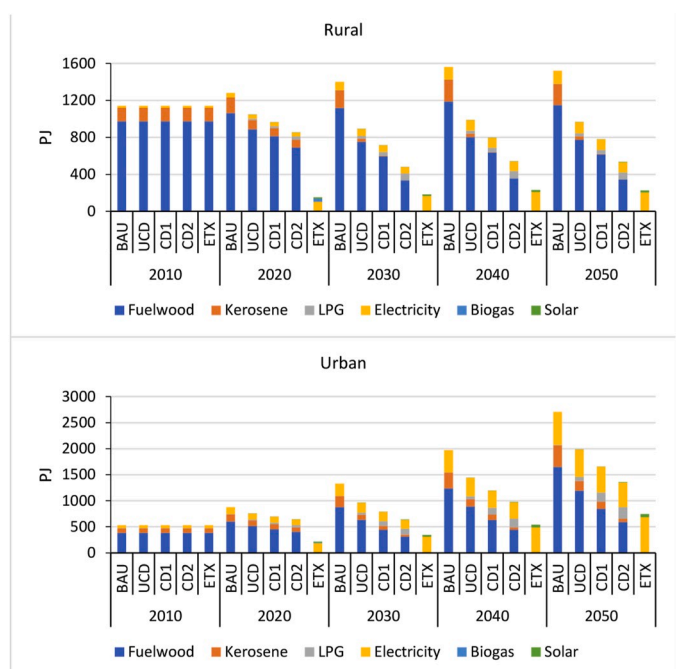


Fig. 8. Residential energy demand projections by scenario.

the high demand for personal mobility in the country due to the poor public transport system and the prestige personal vehicles convey. With respect to energy carriers, the 6-fold increase is mainly due to the consumption of petroleum products in the sector (gasoline and diesel). The consumption of gasoline in the sector grows at an alarming rate of over 600% between 2010 and 2050 and accounts for about 73% of the sectoral final demand. The use of CNG and electricity remain insignificant in the sector.

Notwithstanding, in the low-carbon scenarios, we see the final energy demand of the sector falling by 18%, 25%, 36%, and 28% in UCD, CD1, CD2, and ETX scenarios respectively (relative to the BAU scenario). This can be attributed to the efficient means of transportation considered in these scenarios. Importantly, we observe that the consumption of petroleum products reduces by 2050 owing to the introduction of low-carbon fuels like CNG and electricity. Particularly, our analysis indicates that the percentage share of gasoline in the sector's final energy

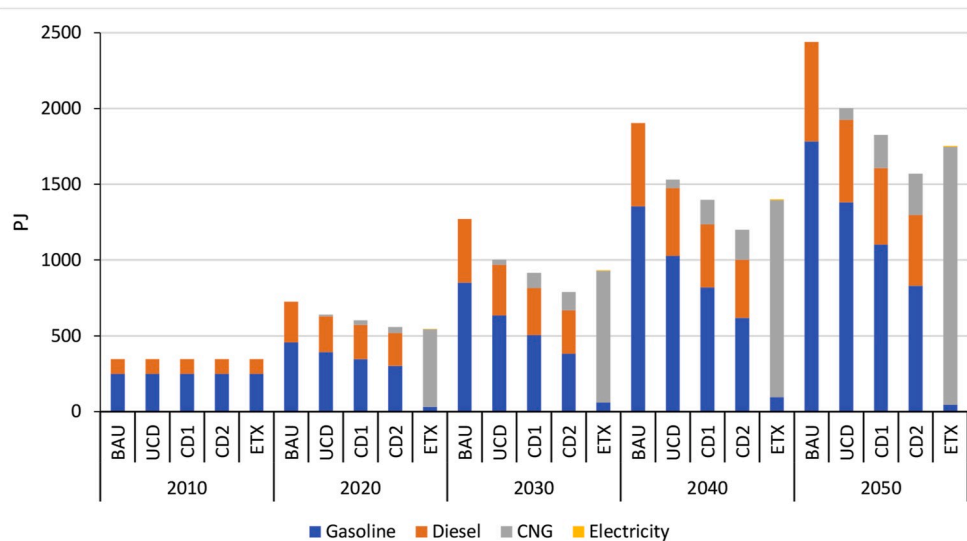


Fig. 9. Transport energy demand projections by scenario.

demand will drop respectively to 69%, 60%, 53%, 3% in UCD, CD1, CD2, and ETX scenarios; values that are much lower than the BAU case. Another interesting aspect of the results is the share of CNG in the ETX scenario. There is a radical transformation in fuel shares in this scenario in the transport fleet. CNG accounts for about 97% of final energy demand in the ETX scenario and this can be attributed to lower cost and carbon content of natural gas in the country when compared to other petroleum products like diesel and gasoline. Additionally, the current result demonstrates how energy consumption trends are not always aligned with CO₂ emissions trends. It may be recalled (Fig. 9) that the ETX scenario (where CO₂ is priced) did not record the lowest final energy demand reduction due to the adoption of some CNG vehicles that are not as efficient as the gasoline type. However, with respect to emissions mitigation, our analysis shows that less carbon fuels were selected in the ETX scenario. This result further implies that efficient technologies alone are not enough to decarbonise the energy system but the type of fuel being used also plays a major role. Thus, decision makers need to fully understand the synergies and trade-offs between energy security and CO₂ emissions mitigation.

4.3.3. Industry sector

Final energy consumption of the industry sector by fuel types for all scenarios are shown in Fig. 10. In the BAU case, industry final energy

demand is expected to grow by a CAGR of 7.7% from 300 PJ in 2010 to around 5857 PJ by 2050—representing a 19-fold increase in 40 years. This amazing growth in final energy demand in the sector is largely on account of the growth in infrastructural demands of the country (such as cement and steel), and the urgent need to improve job opportunities for its teeming unemployed youth. Fuelwood accounts for over 75% of the sector's energy demand and its consumption increases by around 17 times within the modelling timeframe. Natural gas and fuel oil respectively, account for about 22% and 3% of the sector's final energy demand while the use of other petroleum products and coal remain insignificant in the sector. Fuelwood and natural gas are also used to drive thermal power plants to generate electricity in this sector.

With the various measures taken in the mitigation scenarios to help reduce specific energy consumption in industrial processes, the sector's final energy demand is seen to reduce by 20%, 30%, 32%, and 34.5% in UCD, CD1, CD2, and ETX scenarios respectively (relative to the BAU scenario). Our analysis also shows that the mitigation scenarios exhibit some differences in percentage share of fuels when compared to the BAU scenario. The results indicate that the share of natural gas and fuelwood will increase and decrease respectively across the mitigation scenarios. Fuelwood is seen to account for around 70–74% and natural gas is seen to account for around 17–24% of total final energy demand of the sector in the mitigation scenarios. The dominance of fuelwood in the Nigerian

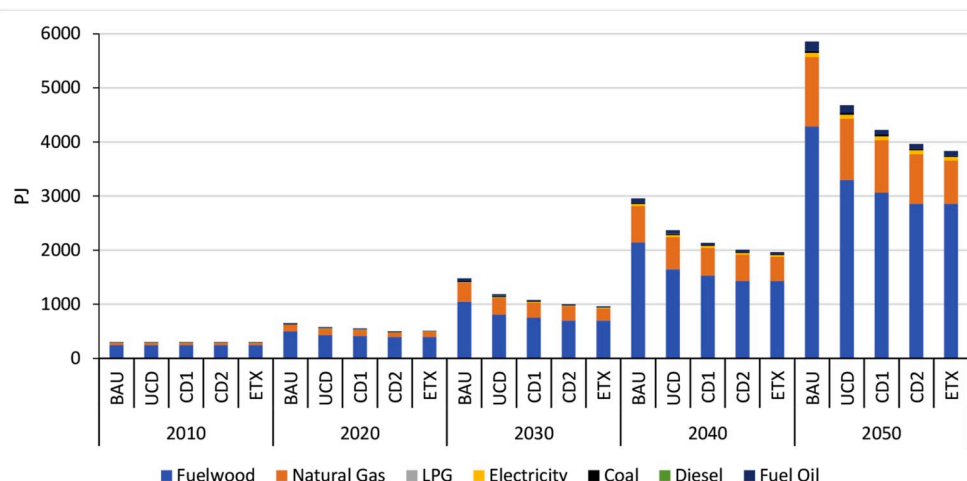


Fig. 10. Industry energy demand projections by scenario.

industry sector is due to the fragmented nature of the sector. The fragmented industries constitute a major part of the Nigerian industry sector and they consume a large amount of fuelwood. However, due to the paucity of data for these fragmented industries, we could not conduct detailed mitigation analysis for them.

4.3.4. Commercial sector

Final energy demand of the commercial sector across all scenarios is reflected in Fig. 11. The final energy demand of the sector in BAU scenario grows roughly 6 times from 203 PJ in 2010 to 1368 PJ in 2050. Electricity and fuelwood are the two main fuels consumed in the sector with shares of around 74% and 22% respectively in the base year which gradually changed to 94% and 5% respectively by 2050. The massive consumption of electricity in this sector is mainly driven by the demand for cooling (air conditioning) in the sector. However, in the mitigation scenarios, by 2050, the final energy demand of the sector declines respectively, by 16.4%, 22.9%, 28.4%, and 36.8% in UCD, CD1, CD2, and ETX scenarios (relative to BAU scenario). As anticipated, electricity and fuelwood still remain the most dominant fuels in the mitigation scenarios with the exception of the ETX scenario. It should be recalled that in the ETX scenario, the model is allowed to endogenously select the cheapest technologies under a carbon tax. Thus, we see that biogas and electricity are the only consumed fuel in this scenario and the use of electricity grows by around 6 times in 40 years. This is mainly due to the electrification of almost all the sector's end uses in this scenario.

4.4. CO₂ emissions trajectory

Fig. 12 reflects the total energy-related CO₂ emissions for the BAU and low-carbon scenarios. In BAU scenario, emissions rise from 76 MtCO₂ in 2010 to reach 228 and 572 MtCO₂ in 2030 and 2050 respectively. This is equivalent to 6.5 times increase in 40 years. Comparing the emission trajectories in the UCD, CD1, CD2, and ETX scenarios to the BAU scenario shows that emissions will drop by 27%, 37%, 46%, and 77% respectively, by 2050. It is only worthwhile to note how radical emission transformations are in the mitigation scenarios as this will give us a sense on the scale of efforts required in decarbonising the Nigerian energy sector in view of achieving its NDC.

At the sectoral levels, Fig. 13 shows the emission trajectories for the BAU and mitigation scenarios. In BAU, the power sector is the largest contributor to total CO₂ emission in 2050 with a percentage share of about 59%. This is not surprising since the Nigerian power supply system is dominated by fossil fuels with natural gas accounting for around 85% of the central electricity grid supply system (NPBR, 2015). This is

then followed by transport, residential, and industry sectors with percentage shares of approximately 30%, 8%, and 3% respectively. The results also show that energy-related emissions from the commercial and agriculture sectors seem to be insignificant in the context of total CO₂ emission from the Nigerian energy sector. For the commercial sector, marginal emissions are observed from cooking with LPG stove. For agriculture sector, this is because, the sector mostly relies on manual labour with crude and less productive technologies (Akpan and Atan, 2015). In the mitigation scenarios, most of the emission reductions are seen to occur in the power sector. Relative to the BAU scenario, emission reductions of 96, 138, 171, and 338 Mt CO₂ (i.e., reductions of 29%, 41%, 51%, and 100%) are achieved in the power sector in UCD, CD1, CD2, and ETX scenarios respectively. Due to emissions tax, complete decarbonisation of the power sector is observed in the ETX scenario, with emissions in the scenario emanating mainly from only the transport and industry sectors. Per capita CO₂ emissions (Fig. 14) in the BAU scenario, grows from 0.48 tCO₂/person in the base year to about 1.44 tCO₂/person in 2050. However, with emissions countermeasures in the mitigation scenarios, our analysis shows that per capita CO₂ emissions by 2050 in UCD, CD1, CD2, and ETX scenarios will drop to 1.05, 0.90, 0.77, and 0.32 tCO₂/person respectively.

4.5. Annual investment projections

One of the benefits of TIMES model is its ability to provide insights on the level of investment required in achieving an energy scenario. This is generally common with partial equilibrium models as they are mainly driven by macro-economic variables which depict how the economy will unfold over a particular time frame (Chiodi et al., 2013). Here, we delineate how the relationship between the energy system and the economic requirements will evolve during the modelling timeframe. Fig. 15 illustrates the annual investment component for each scenario. The results indicate increasing annual investments over the modelling timeframe for all scenarios at different CAGRs. Investments in the BAU is expected to rise at a CAGR of 5.3% from US\$28 billion in 2010 to around US\$88 and US\$224 billion in 2030 and 2050 respectively. In the mitigation scenarios, by 2030, investment requirements grow respectively, by around 0%, 2%, 3%, and 17% in UCD, CD1, CD2, and ETX scenarios in comparison to BAU. However, by 2050, the investment in the BAU scenario slightly grows above the ones in the mitigation scenario. The results may be surprising; however, this is due to the cost-effective investments in energy efficiency in the mitigation scenarios which in turns reduces the cost of fuel, and in the long run, reduces the overall investment requirements. Thus, the present result points to the fact that

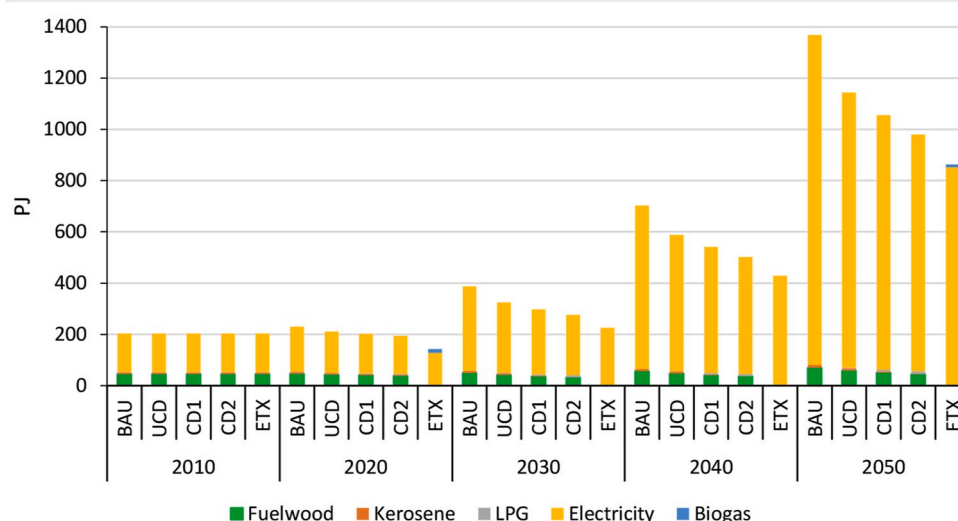


Fig. 11. Commercial energy demand projections by scenario.

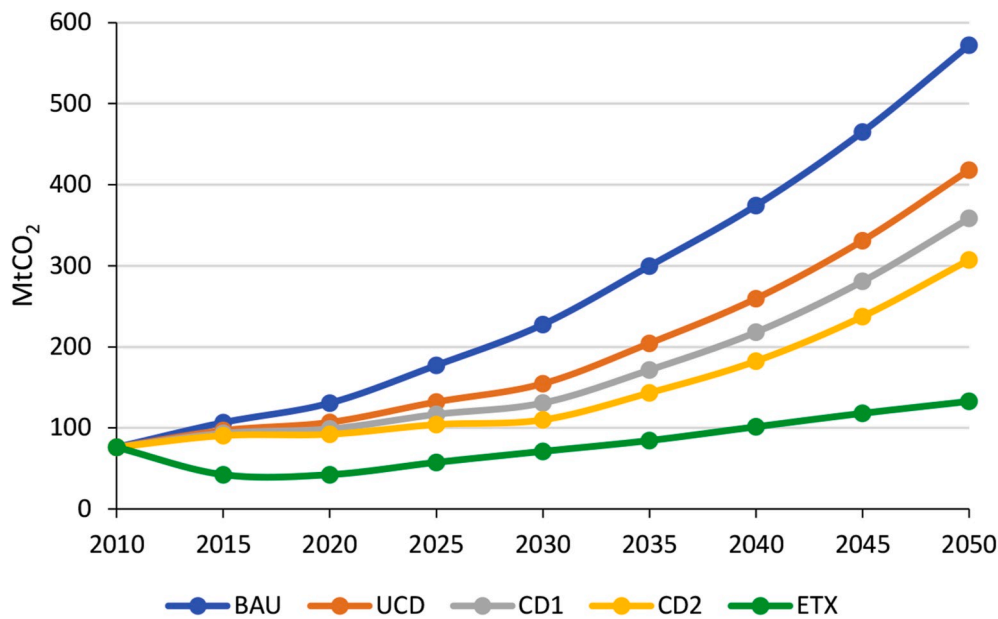
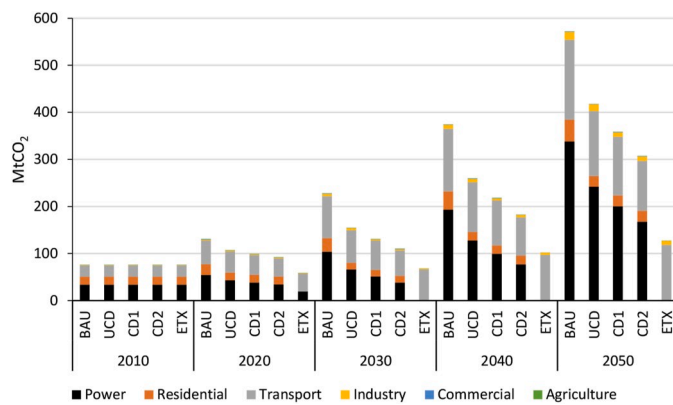
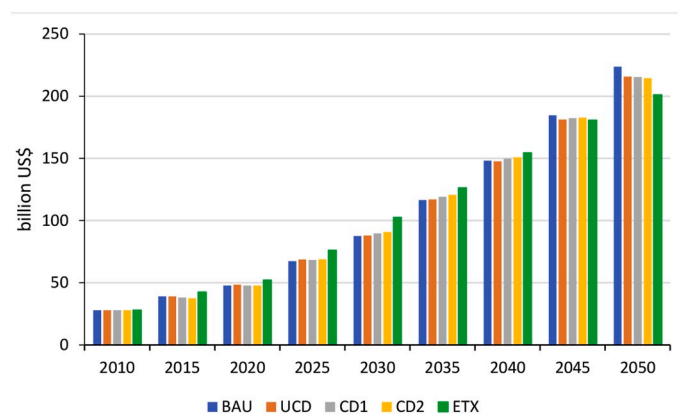
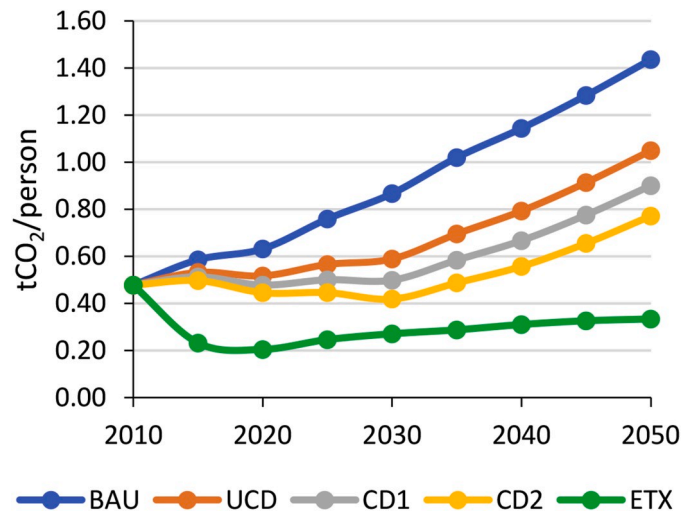
Fig. 12. Total CO₂ emissions projections by scenario.Fig. 13. Sectoral decomposition of total CO₂ emissions.

Fig. 15. Projections of annual investments by scenario.

Fig. 14. Projections of per capita CO₂ emissions by scenario.

investments in climate mitigation are actually cheaper in the long-run when analysed holistically. However, it is worth mentioning that this does not ignore the fact that the initial capital outlay of energy-efficient technologies is significantly higher than the inefficient ones.

4.6. Co-benefits of mitigation scenarios

The results of the energy security benefits of the BAU and mitigation scenarios are presented in Figs. 16 and 17. Lower values of EI indicates that energy security is improving and vice versa. For all scenarios, it can be seen that there are definitive improvements in terms of EI (Fig. 16). This is because of the gradual energy efficiency measures deployed in demand-side technologies. This also illustrates the gradual decoupling between economic growth and energy consumption. However, our analysis suggests that the mitigation scenarios deliver better EIs at different rates when compared to the BAU scenario. The ETX scenario delivers the highest energy security benefits by 2050. With respect to ED, the larger values of ED index signify better energy diversity and thus, improved energy security. Our analysis (Fig. 17) shows that the diversification of energy supply increases in the mitigation scenarios up to 2030 and slightly dropped in 2050. The highest ED values are attained in the ETX scenario both in 2030 and 2050, as they are considerably higher than the one in BAU scenario. In the base year, we observe

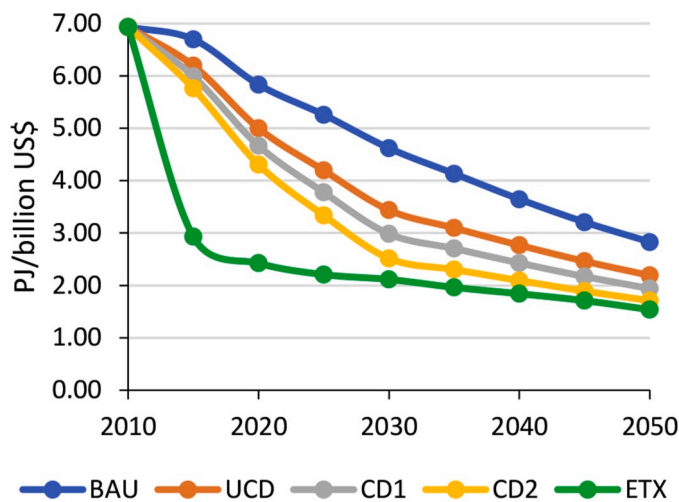


Fig. 16. Projections of energy intensities by scenario.

limited ED (44.2%) due to the low penetration of sustainable energy sources in the supply mix (Authors' calculations). However, this value increased to around 52–66% in the mitigation scenarios; a trend which needs to be improved and thus calls for more aggressive development of sustainable energy resources in the country.

However, it is worthwhile to state that while the results are in numbers, it is essential that their implications in terms of co-benefits are not lost. It can be observed that the two indicators used in this paper to evaluate energy security show meaningful positive improvements in the mitigation scenarios. Accordingly, these co-benefits should be taken into account when developing low-carbon policies for a country like Nigeria as these are essential in achieving other socio-economic goals.

5. Conclusions and policy implications

5.1. Conclusions

The long-term energy and carbon emissions trajectories of Nigeria will be determined by the many choices Nigeria makes today. This paper presents a preliminary model-based analysis of the energy system implications of Nigeria climate mitigation strategies. The analysis presented here indicates significant potential for Nigeria to reduce fossil fuel in its energy supply mix, reduce growth in energy consumption, as well as reduce CO₂ emissions by implementing the mitigation strategies

in its NDC. The results show that final energy demand reductions of 22.4%, 31.3%, 39.5%, and 45.5% and CO₂ emissions mitigation of around 27%, 37%, 46%, and 77% are achievable in UCD, CD1, CD2, and ETX scenarios respectively, by 2050. The results also show that the household sector will be the major sector for reducing energy consumption while the power sector will be the focus for emissions reduction. Therefore, there is a need to promote mitigation actions in the sector (power) in which the profound effect can be observed. The study also showed that there are long-term economic advantages from climate mitigation, and this will also deliver other desirable benefits like improved energy security. However, to realise the full benefits of the Nigerian NDC mitigation plans, there will be a need for robust policy implementation strategies from local to federal government levels. The analysis reported in this study is expected to not only provide fundamental insights into Nigeria NDC but also to provide useful information to inform longer-term low-carbon development policies in the country.

5.2. Policy implications

The study has a range of implications for Nigeria and the international community. At the domestic front, the study has implications for ordinary Nigerian citizens, companies, and policymakers. First, Nigerians need to start adopting energy-efficient appliances and renewable energy technologies (e.g. efficient refrigerators and solar water heaters) at the household level. While the life-cycle costs of these appliances are lower in comparison to the inefficient ones, their initial capital outlay is higher. This poses a challenge for energy-efficient technologies diffusion in the country, especially for low-income households. Thus, a switch to efficient technologies will require the participation of the Nigerian government in terms of policy incentives such as targeted subsidies and micro-finance grants for efficient appliances and renewable energy technologies. Similar actions can be extended to other sectors such as transport and industry. The government can subsidise the prices of cleaner transport fuels (CNG and biofuels) as well as reduce the import duty on efficient vehicles—these incentives would encourage Nigerians to adopt cleaner transport systems.

The study also shows that the switch to renewable energy technologies (mainly decentralised solar PV) will facilitate the decarbonisation of the Nigerian power sector (Fig. 13). This implies that power generation and distribution companies need to start adjusting their portfolio of energy sources. Policymakers need to establish policies that will mandate the power companies to invest in solar PV such as the Renewable Energy Portfolio Standard which has already been adopted in many countries like USA, Italy, Belgium, United Kingdom etc. Moreover, the quest for solar PV in the country also challenges Nigerian

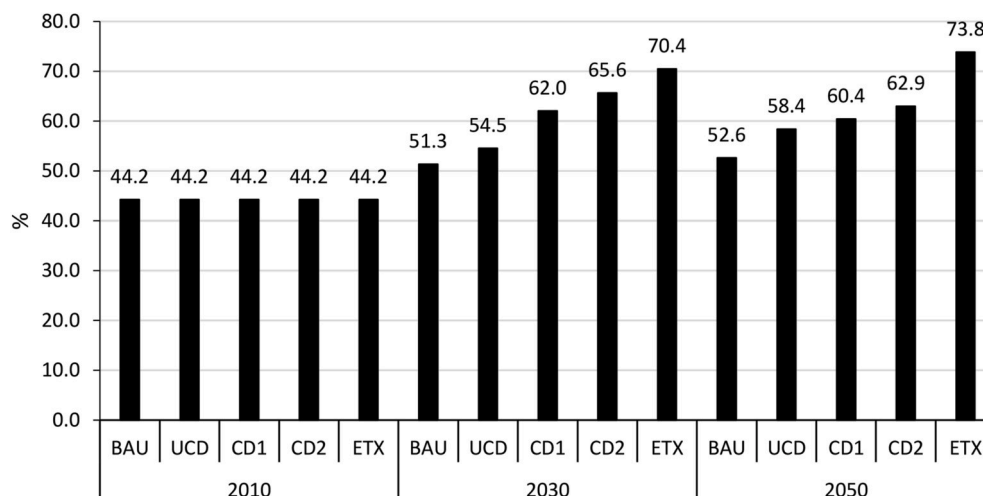


Fig. 17. Energy diversity indices by scenario.

policymakers to strengthen the capacity of the National Agency for Science and Engineering Infrastructure (NASENI), to start manufacturing solar PV panels locally, as this will reduce the cost of importation and thus, encourage power companies to invest in the technology. Currently, Nigeria has no nuclear power plant. However, considering the cost of CO₂ emissions as in the ETX scenario, it is advisable for the country to tap into the local available nuclear resources as this is one of the cheapest options for low-carbon power system. However, safety remains an issue in which policymakers need to pay serious attention to with respect to this technology. Thus, international support will be needed in developing this technology in the country as well as reducing the perceived politics involved in developing nuclear power generation.

The issue of climate change is universal, and the efforts to combat climate change should be considered as a global fight which requires cooperation among countries and institutions. The advancement of knowledge frontiers on which the future energy technology transitions occur will depend on global efforts (Shukla et al., 2008). Our study shows that Nigeria can greatly decarbonise its energy system (by 37% and 46% respectively in 2050) as seen in the CD1 and CD2 scenarios with international support in the form of financial mobilisation and technology transfer. Under the Paris Agreement, around US\$100 billion annually from 2020 was jointly agreed to be used to finance climate mitigation and adaptation capacities in developing countries.⁵ A significant share of this amount will be coming from the Global Climate Fund (GCF) of the UNFCCC. Thus, the international community need to support Nigeria to build its own capacity to access this fund in order to achieve more ambitious mitigation targets as embedded in CD1 and CD2 scenarios. Beyond the GCF, international support is also needed to access other climate funds such as the Clean Investment Funds of the World Bank and the Global Environment Facility. There is also a need for innovative domestic funds mobilisation. For example, the Green Bonds of the Federal Ministry of Environment need to be promoted.⁶ As an oil rich country, the government can put an emission levy on the multinational oil companies. The government also need to use a wide range of its financing instruments such as equity, investment risks and debt relief packages to attract significant doses of private sector funds towards low carbon projects. Hence, the Nigerian government need to establish a robust climate finance framework as a way to blend the multiple sources of both public and private funds together to support climate change initiatives.

With regards to technology transfer, Nigeria needs to look within its national circumstances to identify the best framework. Generally, all stakeholders in the country need to be consulted to conduct a technology needs assessment and the financial implications. For now, decentralised solar PV seems to be the best appropriate technology for Nigeria in terms of rapidly upscaling rural electricity access without compromising environmental sustainability. Then this information will be used to assess whether there is an enabling environment for the technology to thrive in Nigeria. Next, local capacity need to be built around these range of technologies. This is necessary because local manpower need to be trained to install, operate, and maintain technologies to adapt to the prevailing Nigerian circumstances. Finally a mechanism for transfer of the technology will be facilitated and this will also involve bringing down the cost of the technologies.

Additionally, our analysis indicates that a carbon tax can decarbonise the Nigerian energy system by around 77% (ETX scenario) which is very significant in comparison to other scenarios. This suggests that a carbon tax can be an effective instrument for achieving the Nigerian NDC. However, in a globalising world, a single country cannot decide on a development pathway that is very different from the global trend. The

carbon tax used in this study is based on the equalisation of carbon prices across nations towards achieving the Paris Agreement (Carbon Market Watch, 2017). Presently, Nigeria has no carbon tax policy and there has been no proposal for such yet. It is “crystal clear” that Nigeria cannot implement a carbon tax of around US\$40–80/tCO₂ by 2020 and US\$50–100/tCO₂ by 2030, because of the current level of economic development in the country. Thus, a significantly lower carbon tax is needed in Nigeria in the future, while other developed countries should compensate for this by making more ambitious carbon tax schemes in order to achieve the same cumulative mitigation required for the Paris Agreement and beyond. This further implies that a globally efficient low-carbon transition would require harmonisation of development policies across countries while respecting their individual national circumstances.

5.3. Future research

This study is not without caveats and this indicates new areas for further research. Firstly, the results presented here are based on a single projection of macro-economic and demographic variables which by themselves are uncertain. Future research may explore the impact of changes in macro-economic variables on the results. Recently, there has been a call to mainstream climate policies into the SDGs. Consequently, we recommend that future studies explore the interplay between climate mitigation and development using bigger integrated assessment models in order to understand the impacts of the mitigation scenarios on food security, health, land-use, water etc. This type of study is needed to inform complementary climate-development policies. Finally, future research should analyse strategies for reducing emissions from gas flaring in Nigeria.

Declaration of competing interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

CRediT authorship contribution statement

Michael O. Dioha: Writing - original draft. **Atul Kumar:** Conceptualization.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2020.111703>.

⁵ <https://unfccc.int/topics/climate-finance/the-big-picture/climate-finance-in-the-negotiations>.

⁶ <https://environment.gov.ng/green-bonds/>.

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